

Sales Performance Analysis using AdventureWorks2019

Documentation Report

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Introduction

AdventureWorks is a fictional manufacturing company. This project analyzes sales performance using SQL Server, Python, and Power BI to transform transactional ERP data into business insights.

Project Planning

Goal: Build an interactive BI solution that helps management monitor sales performance and make data-driven decisions.

Problem Statement

The company has large amounts of transactional data but lacks an analytical dashboard for monitoring revenue, products, customers, territories, and sales trends.

Business Objectives

- Clean and validate data.
- Perform SQL analysis.
- Conduct exploratory data analysis (EDA) using Python.
- Build an interactive Power BI dashboard.
- Provide actionable KPIs for decision-makers.

Scope

The project focuses on Internet Sales data from AdventureWorks2019, covering sales, customers, products, territories, and time analysis.

Stakeholder Analysis

Stakeholder	Needs	Dashboard Benefits
Executive Management	Revenue & profitability	Executive KPIs and trends
Sales Manager	Regional performance	Territory and product analysis
Marketing	Customer insights	Segmentation and category analysis
Finance	Revenue tracking	Financial KPIs

Functional Requirements

- Display KPIs
- Filter by Date, Product and Territory
- Show Top Products
- Customer Analysis
- Year-over-Year Comparison

Non-functional Requirements

- Fast performance
- Interactive visuals
- Scalable model
- Consistent UI

Database Design

Database: AdventureWorks2019

Main fact table: FactInternetSales

Dimension tables: DimCustomer, DimProduct, DimProductCategory, DimProductSubcategory, DimDate, DimSalesTerritory.

A Star Schema was used to optimize analytical queries.

Data Dictionary

Column	Description
SalesAmount	Revenue
OrderQuantity	Quantity Sold
CustomerKey	Customer Reference
ProductKey	Product Reference
OrderDateKey	Date Reference

ETL Process

SQL Server → Data Validation → Python EDA → Power BI Data Model → Interactive Dashboard

Dashboard Planning

Page 1: Executive Overview

Page 2: Product Performance

Page 3: Customer & Territory Analysis

UI/UX Design

The dashboard follows a clean business theme with KPI cards, slicers, and interactive charts. A blue primary color with orange highlights and light backgrounds improves readability.

KPIs

- Revenue
- Orders
- Customers
- Average Order Value
- Quantity Sold
- Top Product
- Top Territory
- Growth Rate

Testing

- SQL validation
- Missing values check
- Duplicate check
- Power BI validation
- Cross-check results

Risks

Potential risks include data inconsistency, missing records, and dashboard performance issues. These were mitigated through data validation, optimized SQL queries, and a star schema.

Future Improvements

- Sales Forecasting
- Machine Learning
- Customer Segmentation
- Real-time Dashboard
- Azure Integration

Conclusion

The project successfully transformed AdventureWorks ERP data into an interactive dashboard using SQL Server, Python, and Power BI, enabling data-driven business decisions through clear KPIs and visual analytics.